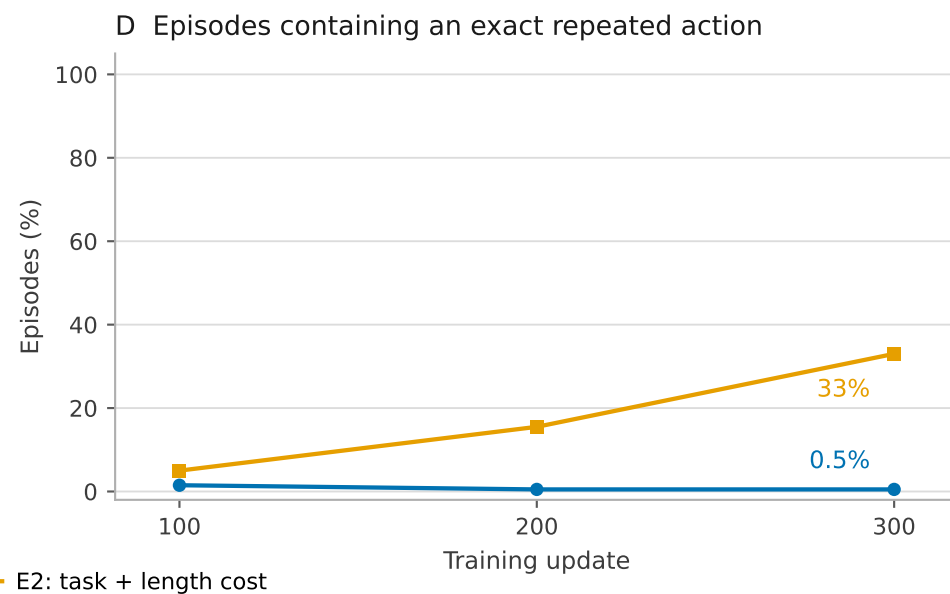
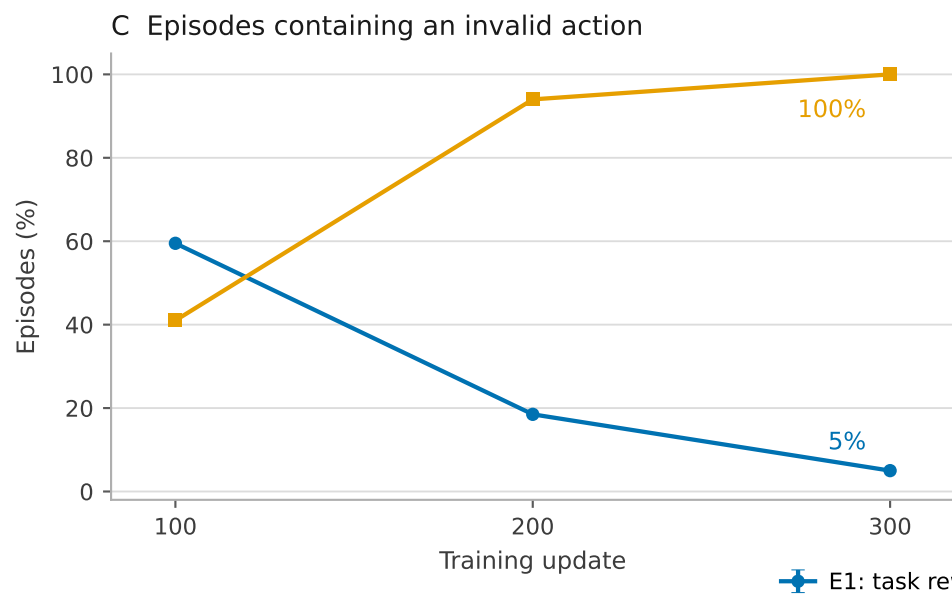
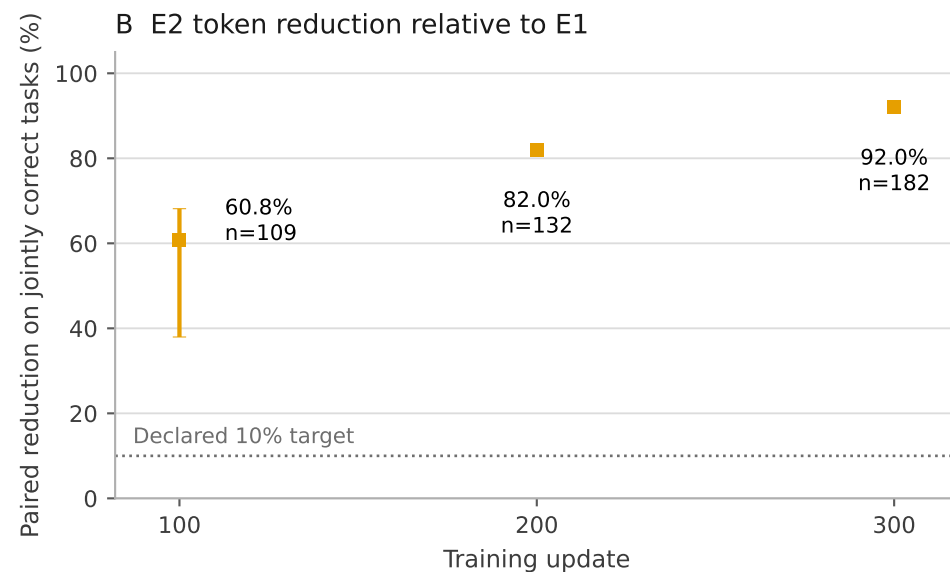
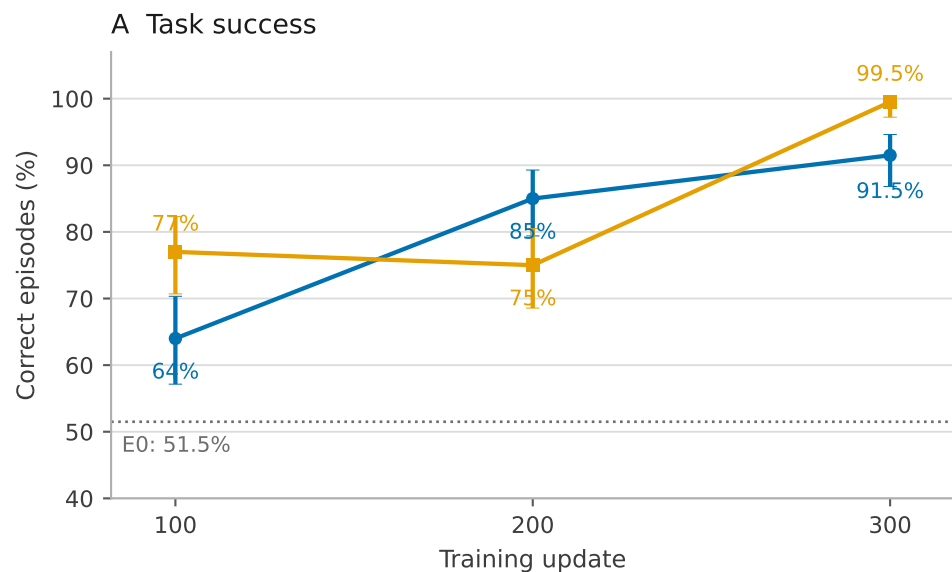


Compression improves as action errors increase

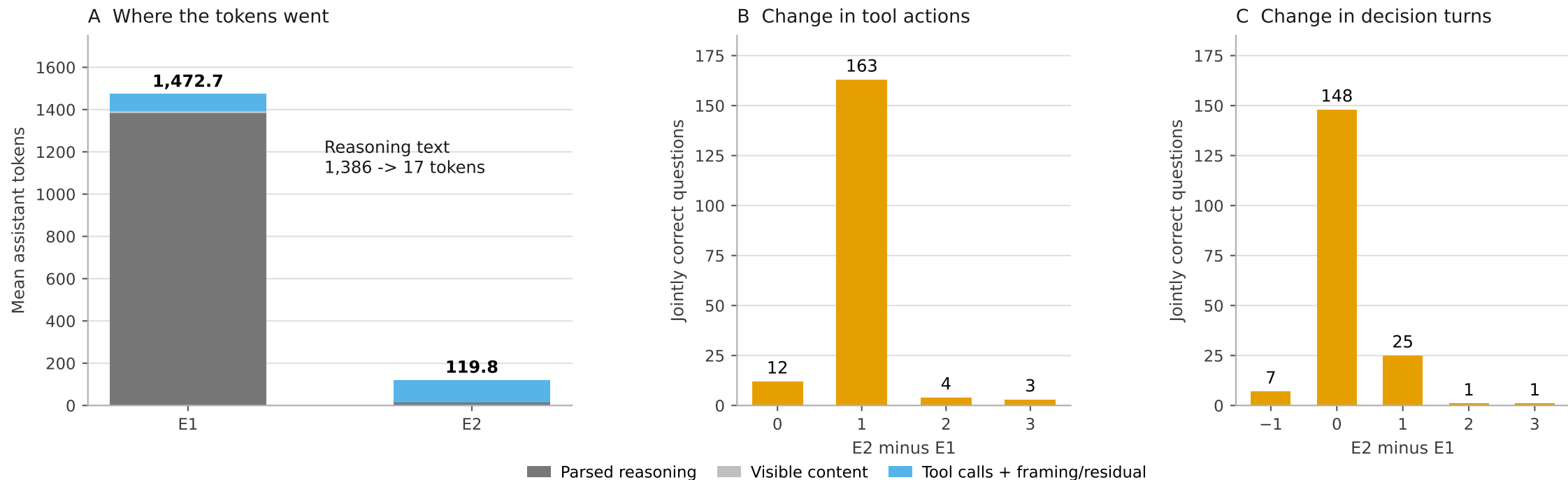
Same 200 held-out inbox tasks at every checkpoint | E1 and E2 trained independently from the base



Success: Wilson 95% intervals. Compression: paired median with 95% bootstrap intervals.
Lines connect three measured checkpoints; they do not describe unmeasured updates. One training seed; update 300 is primary.

Token savings come from generated reasoning

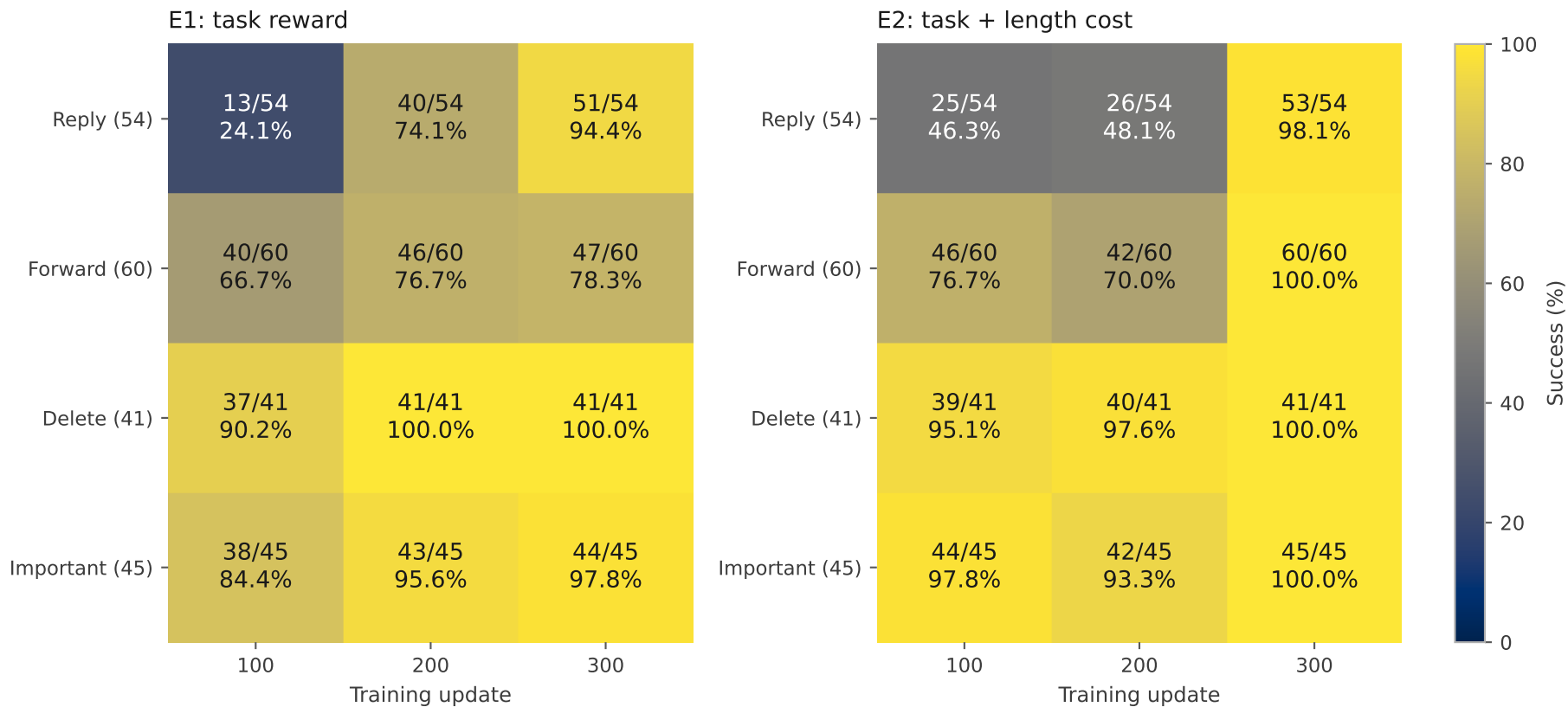
Final checkpoints | The same 182 questions solved correctly by both policies



Parsed reasoning is generated text, not a measurement of internal computation. The residual includes tool calls and template framing.
All 182 E2 responses are shorter. No jointly correct question uses fewer E2 actions; 148/182 use the same number of decision turns.

The temporary quality loss is concentrated in longer workflows

Correct tasks / tasks of that operation | Fixed questions and denominators across checkpoints



Descriptive operation breakdown, not four independent equivalence tests. At update 200, E2 loses 38 E1 successes and gains 18. By update 300, E2 solves 60/60 forwards and 53/54 replies; its remaining failure drops a required period.

A shorter successful trace can contain an avoidable failed action

Final checkpoint, episode 0 | Request: find Floris's email and mark it important

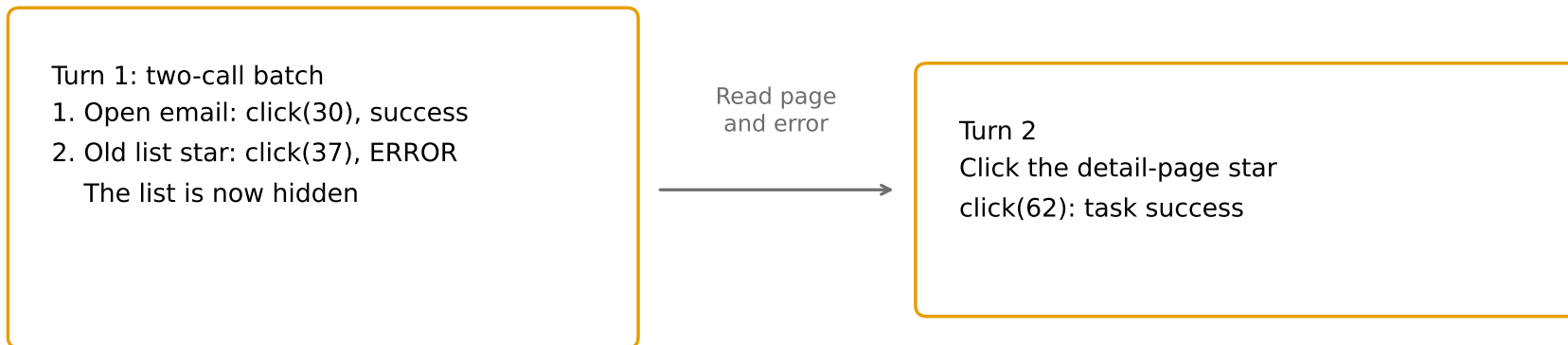
E1

912 tokens
864 reasoning
2 actions



E2

81 tokens
13 reasoning
3 actions

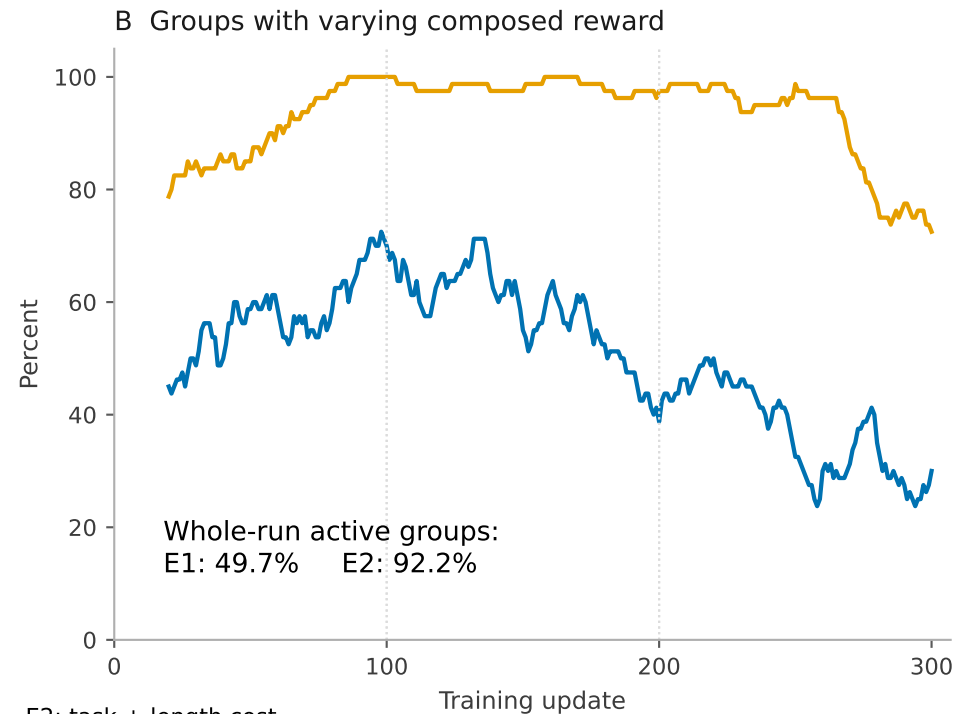
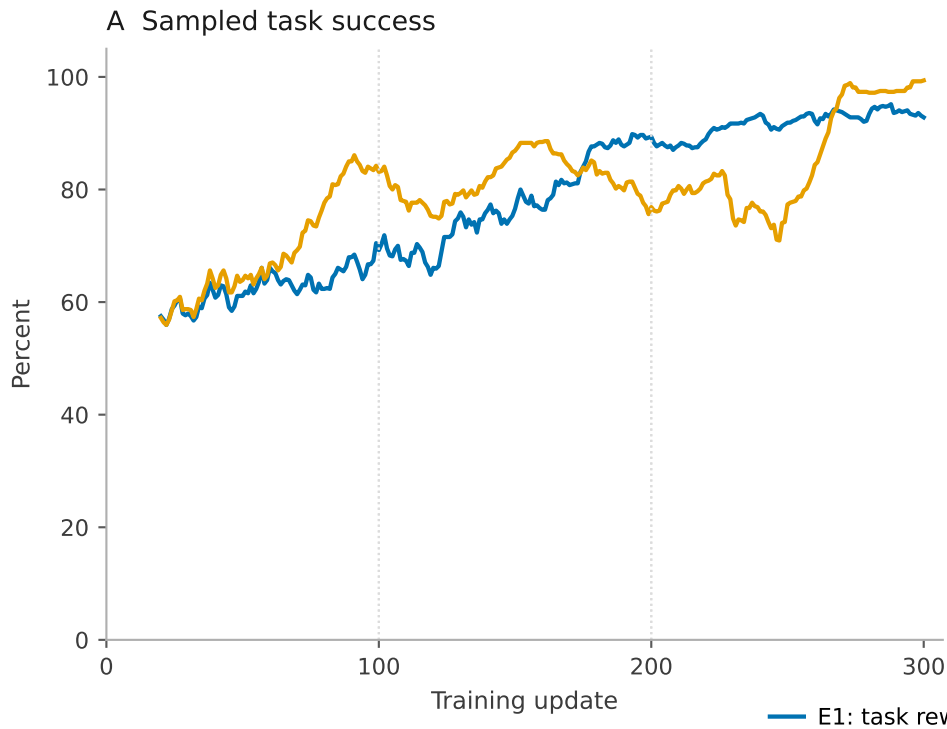


General pattern: all 200 final E2 episodes begin with a valid email-opening click followed by an invalid action in the same batch.

Boxes group calls emitted in one model turn. E2 chooses both first-turn calls before processing either result. The error does not change the saved visible page. This shows an action/state mismatch followed by recovery; intent is unobserved.

Length shaping also changes how often groups provide a learning signal

Training logs | Trailing 20-update means, shown from update 20 | These are sampled training rollouts



Active group = nonzero composed-reward variation ($1 - \text{frac_reward_zero_std}$). It does not quantify gradient magnitude.
This is not held-out success or an independent replication. The current pair cannot isolate length preference from extra gradient availability.